Chemistry and pandemic activity in Laughner et al. (2021)

Status: main text read; supplementary methods unread and requested. This provisional note does not satisfy the requested full-paper review. **Reference.** Laughner et al., *PNAS* 118, e2109481118, doi:10.1073/pnas.2109481118.

Main-text method. The paper synthesizes observations and model experiments on pandemic changes in human activity, emissions, air quality, and greenhouse gases. Its methane example compares deseasonalized TCCON column observations at Park Falls and Lauder with a two-hemisphere box-model scenario. The observational reference is a robust linear trend over 2016–2019; the box model holds methane emissions constant after 2012 and perturbs OH in 2020. Thus it is neither a 2019-baseline flux inversion nor a posterior global methane budget. The broader study also uses MIROC–CHASER assimilation to diagnose ozone production efficiency and GEOS–Chem experiments for nitrate particulate matter.

Main-text findings. The authors report that 2020 methane columns at the two TCCON sites rise about 0.3% above their extrapolated earlier trends. A 3% OH decrease, bracketed by scenarios of 2–4%, yields a similar methane enhancement in the box model. They interpret this agreement as evidence that reduced NO_x emissions and the associated OH decrease could contribute to methane growth. They explicitly allow changes in methane sources as well. The prediction that growth should return toward its earlier rate after NO_x recovers is conditional on OH being the dominant 2020 perturbation; it is not a constraint inferred from 2021–2022 methane observations.

Interpretation for the surge comparison. A scenario selected to explain an enhancement above a 2016–2019 trend cannot be inserted directly as a 2019-based flux anomaly. Doing so would require the box-model initial state, baseline OH and other loss terms, and the time profile of the imposed perturbation. Those details belong to the unavailable supplement. The main-text 3% OH scenario is retained as context, with no numerical conversion to an equivalent emission forcing and no invented posterior uncertainty.

Materials and remaining work. The PMC main article and its figure archive were downloaded. The archive lacks the SI Appendix and Datasets S1–S2. Publisher download attempts return HTTP 403. A search-indexed Caltech author copy could not be retrieved: after diagnosing its certificate-chain problem, the requested file returned HTTP 404. The missing materials were requested from the user. Once available, read the complete supplementary methods and tables, obtain the relevant box-model inputs, and replace this provisional note. No numerical result from the missing supplement is presumed here.